

Total No. of Questions :10]

SEAT No. :

P3235

[5670] 501

[Total No. of Pages :2

B.E. (Civil)

ENVIRONMENTAL ENGINEERING-II
(2015 Pattern) (Semester-I) (EndSem.) (401001)

Time :2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, and Q.9 or Q. 10.
- 2) Figures to the right indicates full marks.
- 3) Draw neat figures wherever necessary.
- 4) Assume any missing data if necessary.
- 5) Use of scientific calculator is allowed.

Q1) a) Explain various factors responsible for self-purification of polluted river. [5]

b) Discuss the effect of change of life style on sewage quality. Also write the typical characteristics of domestic sewage. [5]

OR

Q2) a) Discuss the river classification in per MEF. [5]

b) What is DO deficit? Explain oxygen sag curve. [5]

Q3) a) How the following sewage treatment units helping to treat the waste water?

- i) Screens
- ii) Grit chamber
- iii) Primary sedimentation tank

b) What do you understand by trickling filter? Draw a neat sketch of trickling filter. [2+3]

OR

Q4) a) What is the sludge bulking? Explain the control measures for the same. [2+3]

b) Draw and explain porcess flow diagram for sewage treatment. [2+3]

P.T.O.

- Q5) a)** Discuss the phytoremediation technology for wastewater treatment. Also discuss the advantages and limitations of this process. **[4+2+2]**
- b)** Write working principle: draw a schematic sketch and application of root zone cleaning system for wastewater treatment. **[4+2+2]**

OR

- Q6) a)** Explain the algae bacterial symbiosis in oxidation ponds. Discuss the advantages and limitations of this process. **[4+2+2]**
- b)** Explain the working principle and design criteria of aerated lagoons. **[4+4]**

- Q7) a)** Explain working principle and application of MBR and FMBR. **[4+4]**
- b)** Explain any two methods of sludge disposal with advantages disadvantages and application. **[4+4]**

OR

- Q8) a)** Explain working principle of package sewage treatment plant, write its advantages and disadvantages. **[3+3+2]**
- b)** Draw a neat sketch of up flow anaerobic sludge blanket (UASB) reactor. Explain the principle of working and comment on its suitability for treatment of industrial waste water. **[2+3+3]**

- Q9) a)** Give in tabular form the characteristics of combined effluent from a sugar industry. **[4]**
- b)** Draw and explain units of treating dairy wastewater. **[6]**
- c)** Explain in brief primary and secondary treatment process adopted for treating industrial wastewater. **[4+4]**

OR

- Q10) a)** Explain the principle of working and need of the following treatment units **[9]**
- i)** Equalization
- ii)** Neutralization.
- b)** State the sources and characteristics of Distillery wastewater and draw suitable flow diagram for its treatment. **[9]**

